

AAYUSH NITIN MAHESHWARKAR

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Professional Summary

Senior Software Engineer with 5+ years of experience contributing to large-scale, revenue-generating enterprise platforms. Skilled in Java, Python, Spring Boot, and REST APIs, with a strong track record of cross-functional delivery across global systems. Hands-on experience in production AI instruction design, performance optimization, and production hardening.

Professional Experience

Federal Express Corporation

Senior Software Engineer

Orlando, FL
Jul 2021 - Present

Promotion: Full Stack Developer II (Jul 2021) → Full Stack Developer III (Feb 2023)

- Contributed to Java/Spring Boot rating and pricing platforms handling 2.5M+ peak daily requests, including Quote Platform work supporting \$450M billed and \$1.8B quoted revenue across 69 markets.
- Implemented dangerous-goods surcharge precedence logic in Java and Spring Boot, governing surcharge stacking across country-specific rules and enabling migration of 34,000+ enterprise customers, unlocking \$41M direct and \$1.2B in linked revenue.
- Reduced customer-facing rate-quote API latency by ~600ms by designing proactive Okta JWT pre-caching using Caffeine with async background refresh, eliminating synchronous token-fetch overhead from the pricing API.
- Owled quality standards and final approval authority for Microsoft Copilot LLM instruction sets across multiple enterprise platforms, establishing evaluation criteria for clarity, completeness, security, and accuracy before production release.
- Contributed to a LangChain and Claude-powered agentic pipeline eliminating a 5-day manual error-correction cycle; implemented database-level retry controls and authored Python scripts for bulk data cleanup and revenue recovery.
- Implemented Caffeine Loading Cache across rating services by identifying high-value cache candidates, reducing downstream service calls across 2.5M+ daily requests.
- Improved service reliability by implementing Resilience4j circuit breakers, retry, and fallback handling around downstream API calls, with Log4j structured logging for production diagnostics.

Selected Work Projects

Agentic AI Error-Correction Pipeline - LangChain + Claude

- Contributed to a LangChain and Claude-powered agentic pipeline that autonomously triages and remediates functional rating failures before invoicing resubmission, built for a projected 10x surge from 2,000 to 20,000 daily shipments, eliminating a 5-day manual remediation cycle.
- Implemented database-level retry controls to prevent infinite loops and authored Python automation for bulk data cleanup, accelerating recovery of previously uninvoiced revenue.

Rate Quote API Latency Optimization - Caffeine + Okta JWT

- Diagnosed synchronous Okta JWT token-fetch as the dominant bottleneck on the customer-facing rate-quote API and implemented Caffeine refreshAfterWrite pre-caching with async background refresh, reducing latency by ~600ms; pattern adopted by adjacent teams.

IBM India Pvt. Ltd.

Software Engineer Intern

Bangalore, India
Jan 2019 - Jul 2019

- Performed performance and capacity planning for CICS Transaction Gateway v9.2 using Z/OS, JCL, Java, and REST/JSON services, identifying and resolving production bottlenecks across enterprise transaction workloads.

Technical Skills

Languages: Java, Python, SQL, Shell Scripting, C

Backend & Microservices: Spring Boot, Spring Cloud, Spring Security, Hibernate/JPA, REST APIs, Feign, JMS Messaging, Service Discovery (Eureka), Cloud-Native Architecture, Event-driven Architecture

Cloud & DevOps: Docker, Jenkins CI/CD, Gradle, Linux, Git

Security: OAuth 2.0, SSO, Okta, JWT, Spring Security

Databases & Caching: PostgreSQL, Oracle, MySQL, MongoDB, Redis, Caffeine (Loading Cache, refreshAfterWrite)

Observability: Log4j, Resilience4j (Circuit Breakers, Retry), Splunk, Dynatrace, Grafana, AppDynamics

AI & LLMs: LangChain, Claude, Microsoft Copilot, LLM instruction engineering, prompt evaluation, RAG, AI governance

Practices: Agile (Scrum, Kanban), Unit Testing (JUnit), Code Reviews, Production Incident Triage

Achievements

- FedEx Hall of Fame Award (FY24):** Recognized for contributions to the Quote Platform initiative - \$450M billed / \$1.8B quoted across 69 global markets in FY23.
- AAAI 2020 Fall Symposium on AI for Social Good (Co-Author):** "Socioeconomic and Geographic Variations Impacting the Spread of Malaria." [\[GitHub\]](#)

Education

M.S. Information Technology

Rensselaer Polytechnic Institute | CGPA 3.8/4.0

Troy, NY
Aug 2019 - Dec 2020

B.Tech Electronics & Communication Engineering (Minor: Computer Science)

PES University | CGPA 7.63/10 | Minor CGPA 8.95/10

Bengaluru, India
Aug 2015 - Jul 2019